

Distributed Computing (Subject Code: 52771)

Module-Wise Question Bank

BE Comp | Sem VIII | CBCGS R-19 | C Scheme

Note: Questions highlighted in red frequency (>1) are repeated across multiple papers.

Module 1 – Introduction to Distributed Systems

Question	Frequency
✓ What are different issues and goals related to distributed systems?	3
✓ Differentiate between NOS, DOS and Middleware in the design of distributed systems.	2
✓ Compare NOS and DOS.	1
✓ Discuss design issues in distributed systems.	1
✓ Enlist the issues in designing the Distributed System. Explain failure transparency & location transparency in detail.	1
✓ Explain issues in designing Distributed system.	1
✓ What are the services offered by middleware?	1

Module 2 – Communication

Question	Frequency
✓ What is RPC? Explain working of RPC in detail.	3
✓ What is Remote Procedure Call? Explain how transparency is achieved in RPC.	1
Differentiate between RMI and RPC.	1
Explain various forms of message oriented communication with suitable example.	1
✓ What is 1:M and M:1 group communication?	1
✓ Explain group communication.	1
✓ Explain Absolute ordering and Casual ordering process with the help of example for many to many communication.	1
✓ Explain the various ordered semantics used for Many to Many communication.	1

Module 3 – Synchronization

Question	Frequency
✓ Explain Raymond's algorithm for mutual exclusion.	4
✓ Explain Bully election algorithm with suitable example.	3
✓ Explain bully election algorithm.	1
✓ What are physical clocks? Explain any one physical clock synchronization algorithm.	2

Question	Frequency
✓ How does Lamport synchronize logical clock? Explain with example. Which events are said to be concurrent in Lamport's timestamp?	2
✓ What is logical clock? Why are logical clocks required in distributed systems? How does Lamport synchronize logical clock? Which events are said to be concurrent in Lamport's timestamp?	1
✓ Explain Ricart-Agrawala's algorithm and how it optimizes the message overhead in achieving mutual exclusion.	2
✓ Justify how Ricart-Agrawala's algorithm optimized the Message overhead in achieving mutual exclusion.	1
Distributed Mutual Exclusion: explain the requirements and performance measures.	1
✓ Explain the need of election algorithm.	1
✓ Explain Chandy-Misra_Hass Algorithm for distributed deadlock detection.	2
✓ Clock Synchronization: explain physical clock, logical clocks and election algorithms.	1

Module 4 – Resource and Process Management

Question	Frequency
Explain different load estimation and process transfer policies used by load balancing algorithms.	4
Justify how load balancing is useful in distributed system.	1
Explain desirable features of global scheduling algorithm.	1
Explain key features of Global Scheduling algorithm.	2
Explain code migration and its techniques.	2
Discuss code migration in distributed system.	1
Describe code migration issues in detail.	1
What is need of code migration? Explain the role of process to resource and resource to machine binding in code migration.	1
Introduction to Process Management, Process Migration, Code Migration.	1

Module 5 – Replication, Consistency and Fault Tolerance

Question	Frequency
✓ What are different data centric consistency models?	2
✓ Explain any five data centric consistency models.	1
✓ Explain how Monotonic read consistency model is different than Read your Write consistency model.	1
✓ Discuss and differentiate various client consistency models.	2

Question	Frequency
What is fault tolerance? Explain failure models.	3
What is fault tolerance? Describe different types of failure models.	1
Write a short note on Replication and the types of it.	2
Introduction to replication and consistency, Data-Centric and Client-Centric Consistency Models, Replica Management.	1
Fault Tolerance: Introduction, Process resilience, Recovery.	1
Discuss the technique to achieve the Process resilience.	1

Module 6 – Distributed File Systems

Question	Frequency
List desirable features of Distributed File System. How are modifications propagated in file caching schemes?	1
Explain file caching schemes.	2
Desirable features of a good DFS.	1
Discuss Google File System (GFS) as a scalable distributed file system.	1
Explain synchronization in DFS with its challenges.	1
Introduction and features of DFS, File models, File Accessing models, File-Caching Schemes, File Replication. Case Study: Network File System (NFS).	1
Designing Distributed Systems: Google Case Study.	1

Unclassified Questions

(Questions that could not be mapped to a specific module)

Question	Frequency
Explain Distributed Shared Memory: Architecture and design issues.	1